

An introduction to TD-DFT

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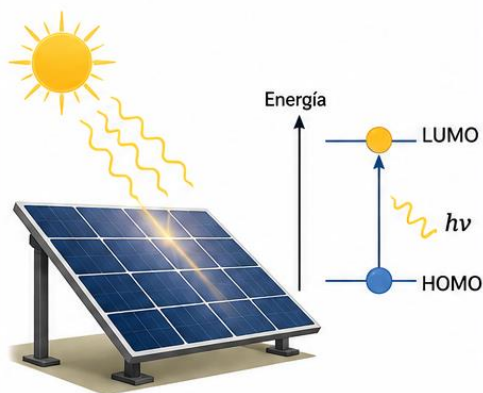


¿Por qué es importante la física de estados excitados?

Los estados excitados gobiernan cómo la materia **absorbe luz y transforma energía**, procesos esenciales en la naturaleza y en tecnologías clave.

1. Energía solar

En celdas solares, la luz excita electrones en semiconductores, generando separación de carga y corriente eléctrica.

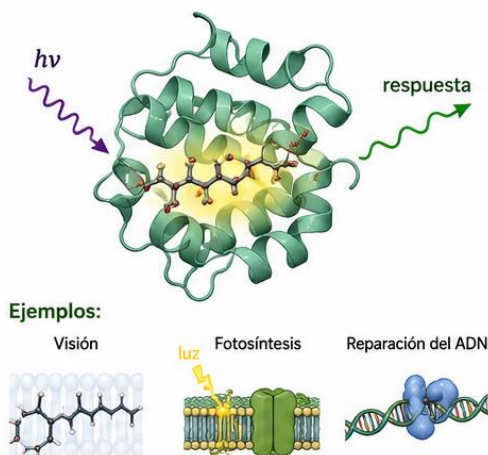


Impacto:

Conversión eficiente de luz solar en electricidad limpia y renovable.

2. Bioquímica y biología

Muchos procesos biológicos comienzan con la excitación electrónica: visión, fotosíntesis, reparación del ADN y más.

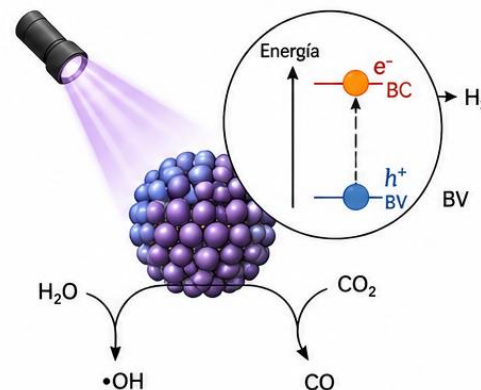


Impacto:

Fundamental para la vida y para el desarrollo de fármacos y terapias.

3. Fotocatálisis

La luz excita catalizadores, generando pares electrón-hueco que impulsan reacciones químicas clave (p. ej., producción de H_2 , reducción de CO_2 , degradación de contaminantes).



Impacto:

Soluciones sostenibles para energía, medio ambiente y síntesis química.



Comprender y controlar los estados excitados permite diseñar materiales y moléculas más eficientes para enfrentar desafíos globales: energía limpia, salud y protección del medio ambiente.

Jablonski Energy Diagram

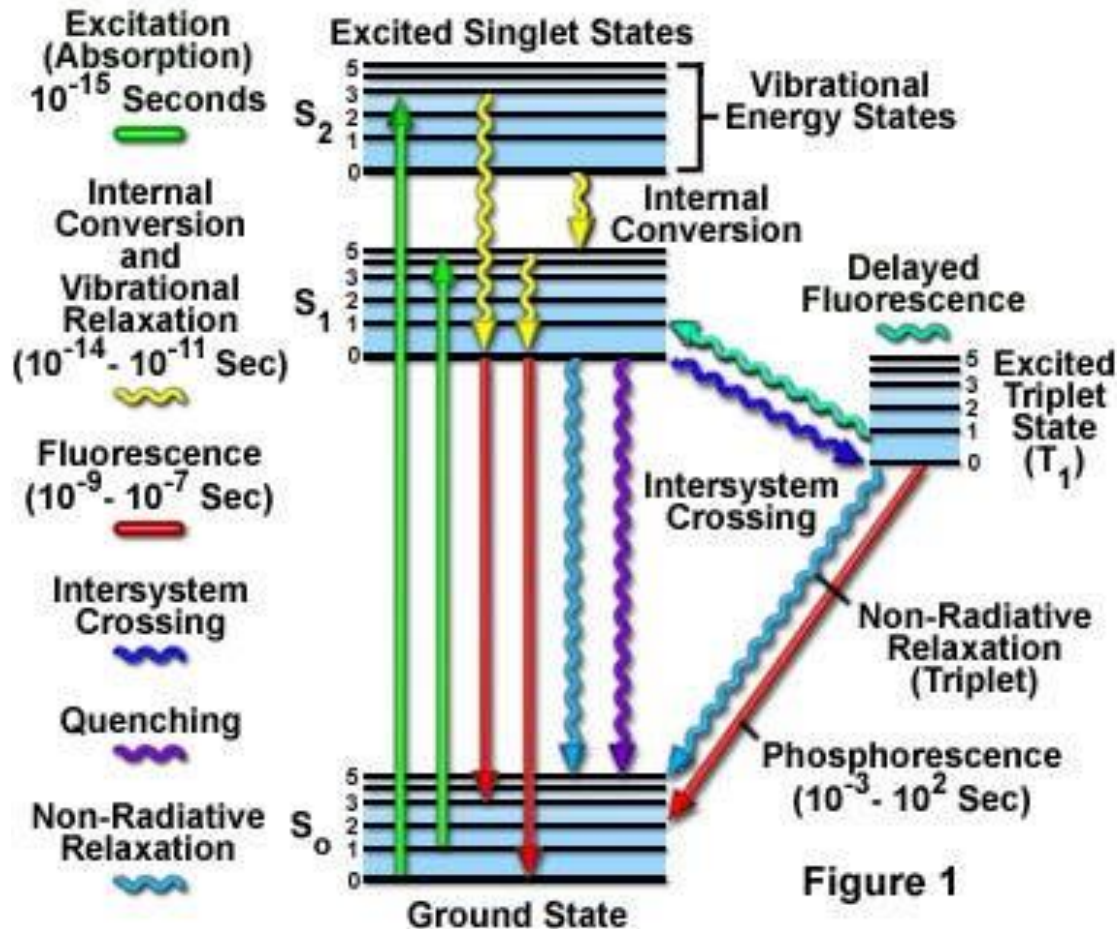


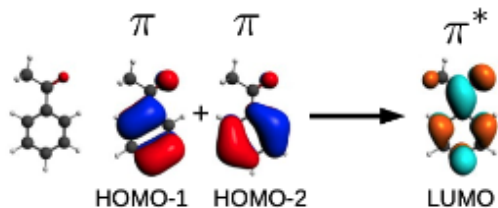
Figure 1

Tipos de estados excitados usuales

Moléculas orgánicas conjugadas :

1. Estados $\pi\pi^*$

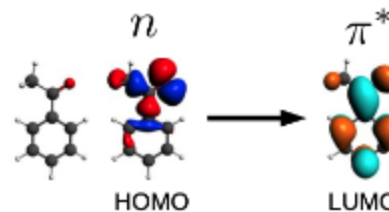
Ej.: Acetofenona



Excitación de un electrón de un orbital π ocupado a un orbital π^* .

2. Estados $n\pi^*$

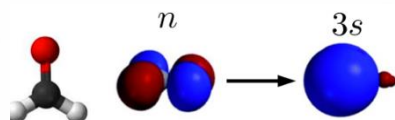
Ej.: Acetofenona



Excitación de un electrón de un orbital no enlazante (n) a un orbital π^* .

3. Estados Rydberg

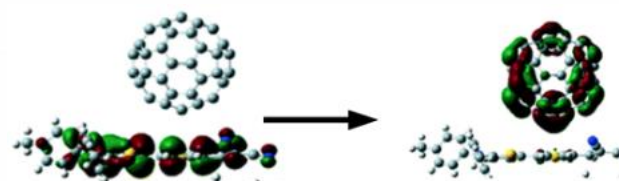
Ej.: Formaldehído



Excitación a un orbital de carácter difuso (tipo Rydberg).

4. Estados de transferencia de carga

Ej.: Fulleren-Tiofeno



Time Dependent Kohn-Sham Equations

Given the 1-1 map between TD-densities and TD-Potential, one can postulate that there exists a non-interacting system (KS system) with initial state $|\Phi_0\rangle$, which yields the right time-dependent density of an interacting system with initial state $|\Psi_0\rangle$. The TD-SE for $|\Phi_0\rangle$ will be

$$i\frac{\partial}{\partial t}|\Phi(t)\rangle = \hat{H}_s(t)|\Phi(t)\rangle \quad t \geq t_0$$

$$|\Phi(t)\rangle = \frac{1}{\sqrt{N!}}|\phi_1\phi_2\dots\phi_N\rangle \text{ Slater's determinant}$$

$$i\frac{\partial\phi_i(\mathbf{r},t)}{\partial t} = -\frac{1}{2}\nabla^2\phi_i(\mathbf{r},t) + v_{KS}[\rho;\Phi_0](\mathbf{r},t)\phi_i(\mathbf{r},t) \quad i=1,2,\dots,N$$

$$\rho(\mathbf{r},t) = \sum_{i=1}^N |\phi_i(\mathbf{r},t)|^2$$

$$v_{KS}[\rho,\Phi_0](\mathbf{r},t) \doteq v_{ext}(\mathbf{r},t) + \int d\mathbf{r}' \frac{\rho(\mathbf{r}',t)}{|\mathbf{r}-\mathbf{r}'|} + \underline{v_{xc}[\rho;\Phi_0;\Psi_0](\mathbf{r},t)}$$

TD-Exchange-correlation potential

$$v_{xc}[\rho; \Phi_0; \Psi_0](\mathbf{r}, t)$$



Initial state of the real system

Initial state of the KS system

Density from t_0 to t

Memory

$$\rho(\mathbf{r}, 0) = \sum_{i=1}^N |\phi_i(\mathbf{r}, 0)|^2 = N \int d\mathbf{r}_2 \cdots \int d\mathbf{r}_N |\Psi_0(\mathbf{x}, \mathbf{x}_2 \dots \mathbf{x}_N)|^2$$

$$\partial_t \rho(\mathbf{r}, 0) = -\nabla \cdot \Im m \sum_{i=1}^N \phi_i^*(\mathbf{r}, 0) \nabla \phi_i(\mathbf{r}, 0)$$

$$= -N \nabla \cdot \Im m \int d\mathbf{r}_2 \cdots \int d\mathbf{r}_N \Psi_0^*(\mathbf{x}, \mathbf{x}_2 \dots \mathbf{x}_N) \nabla \Psi_0(\mathbf{x}, \mathbf{x}_2 \dots \mathbf{x}_N)$$

$$\partial_t \rho(\mathbf{r}, 0) = -\nabla \cdot \mathbf{j}(\mathbf{r}, 0)$$

The fact that the KS potential depends on how Φ_0 is chosen is unique feature in TDDFT

Propagation of Kohn-Sham Equations

$$i \frac{\partial \phi_i(\mathbf{r}, t)}{\partial t} = -\frac{1}{2} \nabla^2 \phi_i(\mathbf{r}, t) + \left(v_{ext}(\mathbf{r}, t) + \int d\mathbf{r}' \frac{\rho(\mathbf{r}', t)}{|\mathbf{r} - \mathbf{r}'|} + v_{xc}(\mathbf{r}, t) \right) \phi_i(\mathbf{r}, t)$$

Time evolution operator:

$$\phi_i(\mathbf{r}, t) = e^{-i \int_{t_0}^t \hat{H}_{KS}(t') dt'} \phi_i(\mathbf{r}, t_0)$$

$$\rho(\mathbf{r}, t) = \sum_{i=1}^N |\phi_i(\mathbf{r}, t)|^2$$

<http://www.tddft.org>



octopus

Is there a variational principle in TD-DFT?

In ground state one uses the minimum energy principle to construct $Exc[\rho]$.

$v_{xc}(r)$ is given just by the functional derivative $\frac{\delta Exc[\rho]}{\delta \rho}$

Be the Dirac-Frenkel action (RG)

$$A[\Psi] = A[\Psi_0, \rho] = \int_{t_0}^{t_1} dt \langle \Psi(t) | i \frac{\partial}{\partial t} - \hat{H}(t) | \Psi(t) \rangle$$

$A[\Psi]$ is variational with respect to the wavefunction $|\Psi(t)\rangle$ that evolves from $|\Psi(t_0)\rangle$ under the action of $\mathbf{H}$.

$$\frac{\delta A[\psi]}{\delta \langle \Psi |} \stackrel{RG}{=} \frac{\delta A[\rho]}{\delta \rho(t)} = 0$$

Which is equivalent to the TD-SE

$$\frac{\delta A[\psi]}{\delta \langle \Psi |} = \left(i \frac{\partial}{\partial t} - \hat{H}(t) \right) |\Psi(t)\rangle = 0 \quad \text{if} \quad \delta |\Psi(t_0)\rangle = \delta |\Psi(t_1)\rangle$$

Causality problem and E_{xc} potential

Be $t > t'$. The action is symmetric with respect to t and t' in the following sense:

$$0 \neq \underbrace{\frac{\delta^2 A[\rho]}{\delta \rho(t') \delta \rho(t)}}_{\text{time ordered}} = \frac{\delta^2 A[\rho]}{\delta \rho(t) \delta \rho(t')} = 0 \quad t > t'$$

This problem is deeply discussed by Robert van Leeuwen (1998).

$$\delta |\Psi(t_0)\rangle = \delta |\Psi(t_1)\rangle \quad \text{But TD-SE is a first order DE with respect to } t, \text{ therefore:} \quad |\Psi(t_0)\rangle \neq |\Psi(t_1)\rangle$$

If one writes the action for the KS system and split V_{ks} as usual, it takes some time to show that:

$$v_{xc}(\mathbf{r}, t) = \frac{\delta A_{xc}[\rho]}{\delta \rho(\mathbf{r}, t)}$$

Adiabatic Approximation

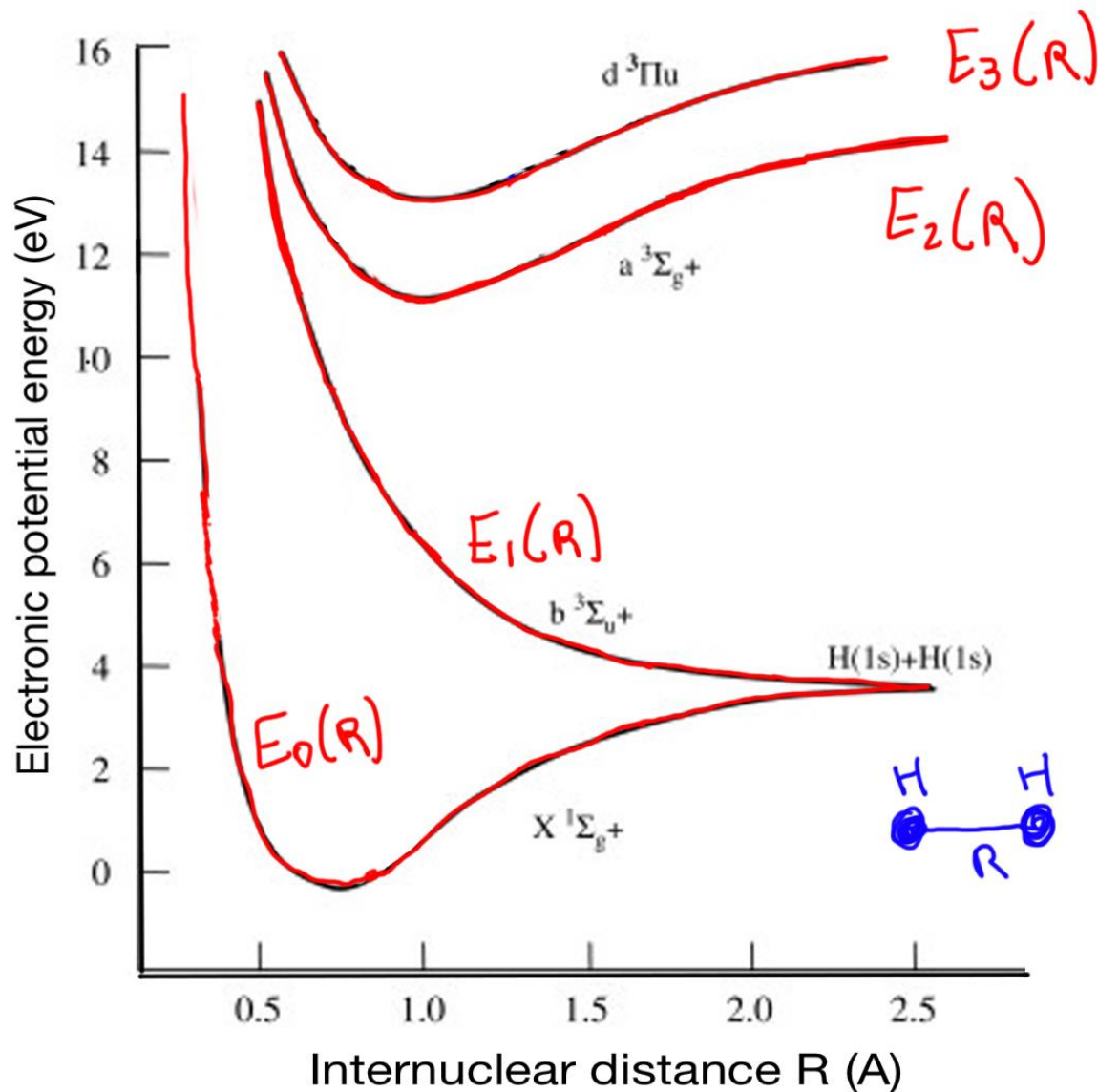
The simplest approximation for A_{xc} is to use XC functional from ground-state (stationary) DFT

ALDA: Adiabatic Local Density Approximation

$$A_{xc}^{ALDA}[\rho] = \int_{t_0}^{t_1} dt \int e_{xc}^{HEG}(\rho(\mathbf{r}, t)) d\mathbf{r}$$

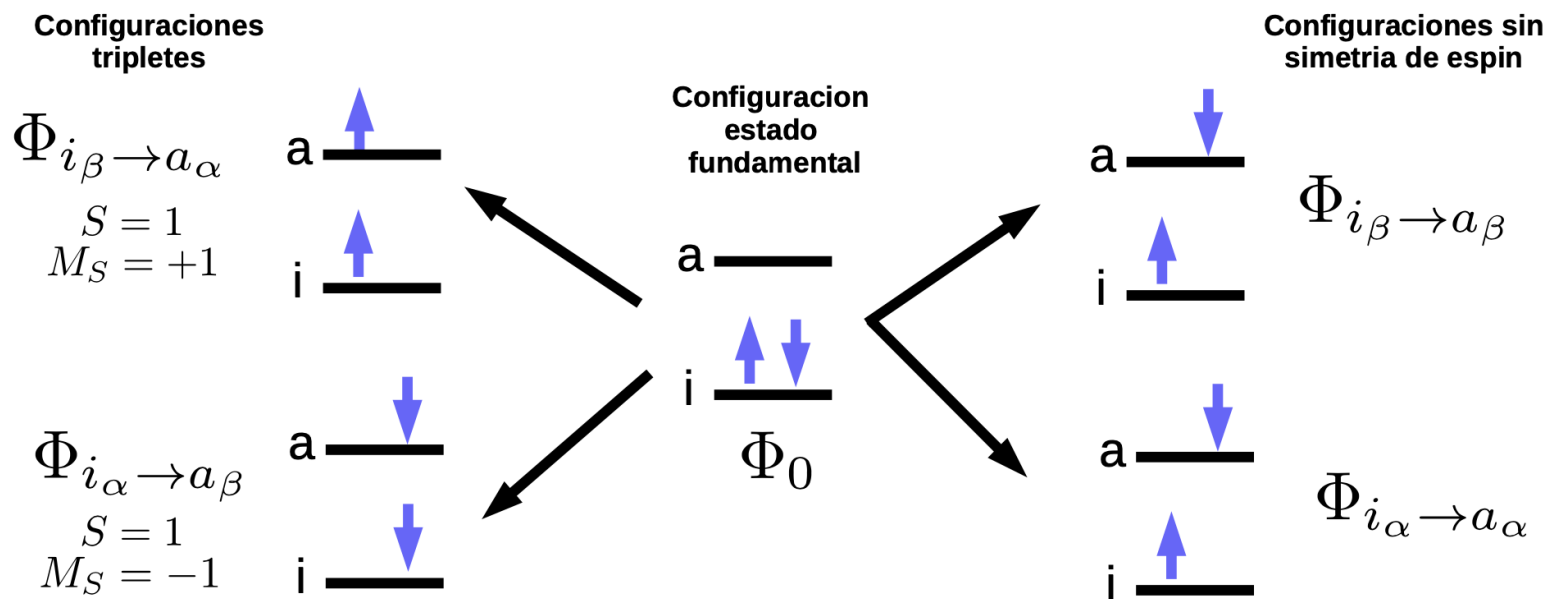
$$v_{xc}^{ALDA} = \frac{\delta e_{xc}^{HEG}(\rho)}{\delta \rho} = v_{xc}^{HEG}$$

Adiabatic potential energy surfaces



Modelo de dos niveles y dos electrones

Del espacio de **excitaciones simples*** obtendremos una base de 4 configuraciones, donde construimos 3 estados tripletes (degenerados) y 1 estado singulete:



*Respecto a la configuración del estado fundamental:

Excitaciones simples: transición de un electron a un orbital virtual

Excitaciones dobles: transición de dos electron a orbitales virtuales

Wavefunctions

$$\Phi(r_1, r_2) \times \Phi_{spin}(\sigma_1, \sigma_2)$$

$$\begin{array}{l} \Phi_{i_\beta \rightarrow a_\alpha} \\ S=1 \\ M_S=+1 \end{array} \quad \begin{array}{c} \text{a} \uparrow \\ \text{i} \uparrow \end{array} \quad \frac{1}{\sqrt{2}}(\psi_i(r_1)\psi_a(r_2) - \psi_i(r_2)\psi_a(r_1))\alpha(\sigma_1)\alpha(\sigma_2)$$

$$\begin{array}{l} \Phi_{i_\alpha \rightarrow a_\beta} \\ S=1 \\ M_S=-1 \end{array} \quad \begin{array}{c} \text{a} \downarrow \\ \text{i} \downarrow \end{array} \quad \frac{1}{\sqrt{2}}(\psi_i(r_1)\psi_a(r_2) - \psi_i(r_2)\psi_a(r_1))\beta(\sigma_1)\beta(\sigma_2)$$

$$\begin{array}{l} \Phi_{i_\beta \rightarrow a_\beta} \\ \Phi_{i_\alpha \rightarrow a_\alpha} \end{array} \quad \left. \begin{array}{c} \text{a} \downarrow \\ \text{i} \uparrow \\ \text{a} \downarrow \\ \text{i} \uparrow \end{array} \right\} \quad \begin{array}{l} \frac{1}{\sqrt{2}}(\psi_i(r_1)\psi_a(r_2) - \psi_i(r_2)\psi_a(r_1))\frac{1}{\sqrt{2}}(\alpha(\sigma_1)\beta(\sigma_2) + \alpha(\sigma_2)\beta(\sigma_1)) \\ \frac{1}{\sqrt{2}}(\psi_i(r_1)\psi_a(r_2) + \psi_i(r_2)\psi_a(r_1))\frac{1}{\sqrt{2}}(\alpha(\sigma_1)\beta(\sigma_2) - \alpha(\sigma_2)\beta(\sigma_1)) \end{array}$$

Excitation energies in HF

$$\Delta E = \langle \Phi_i^a | H | \Phi_i^a \rangle - \langle \Phi_0 | H | \Phi_0 \rangle = \varepsilon_a - \varepsilon_i + \langle ai || ai \rangle$$

$$\Delta E_{triplet} = \varepsilon_a - \varepsilon_i + \langle ai | ai \rangle = \varepsilon_a - \varepsilon_i + J_{ia}$$

$$\Delta E_{singlet} = \varepsilon_a - \varepsilon_i + 2 \langle ai | ia \rangle - \langle ai | ai \rangle = \varepsilon_a - \varepsilon_i + 2K_{ia} - J_{ia}$$



Particle-hole
Binding energy

Linear-response TD-DFT (Casida's)

Be the external potential split in two pieces:

$$v_{ext}(\mathbf{r}, t) = v_0(\mathbf{r}) + v_{appl}(\mathbf{r}, t)$$

And let the density be expanded in a Taylor's series:

$$\rho(\mathbf{r}, t) = \rho_0(\mathbf{r}) + \rho^{(1)}(\mathbf{r}, t) + \rho^{(2)}(\mathbf{r}, t) + \dots$$

$$\rho^{(1)}(\mathbf{r}, t) = \int_{t_0}^t dt \int \left(\frac{\delta \rho(\mathbf{r}, t)}{\delta v(\mathbf{r}', t)} \right)_N \delta v_{appl}(\mathbf{r}', t')$$

$$\rho^{(1)}(\mathbf{r}, t) = \delta \rho(\mathbf{r}, t) = \int_{t_0}^t dt \chi(\mathbf{r}, \mathbf{r}', t - t') \delta v_{appl}(\mathbf{r}', t')$$

Writing the last equation in the energy domain (Fourier transform)

$$\delta \rho(\mathbf{r}, \omega) = \int d\mathbf{r}' \chi(\mathbf{r}, \mathbf{r}', \omega) \delta v_{appl}(\mathbf{r}', \omega)$$

Lehman representation of LRP

$$\delta\rho(\mathbf{r},\omega) = \int d\mathbf{r}' \chi(\mathbf{r},\mathbf{r}',\omega) \delta v_{\text{appl}}(\mathbf{r}',\omega)$$

$$\delta\rho(\omega) = \chi(\omega) \delta v_{\text{appl}}(\omega) \quad \text{notation!}$$

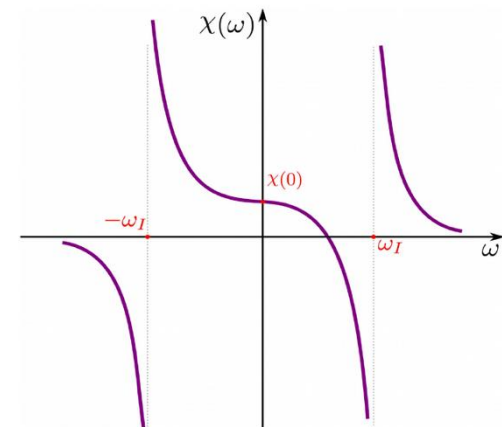
$$\delta v_{\text{appl}}(\omega) = \chi^{-1}(\omega) \delta\rho(\omega)$$

From MBPT one knows that

$$\chi(\mathbf{r},\mathbf{r}',\omega) = \int \frac{\langle \Psi_0 | \hat{\rho} | \Psi_I \rangle \langle \Psi_I | \hat{\rho} | \Psi_0 \rangle}{\omega - (E_I - E_0) + i0^+} - \int \frac{\langle \Psi_I | \hat{\rho} | \Psi_0 \rangle \langle \Psi_0 | \hat{\rho} | \Psi_I \rangle}{\omega + (E_I - E_0) + i0^+}$$

Note that at the poles:

$$\omega_I = E_I - E_0 \rightarrow \chi(\omega) = \infty \rightarrow \delta v_{\text{appl}}(\omega_I) = 0$$



LRP of the KS system

$$\delta\rho(\mathbf{r},\omega) = \int d\mathbf{r}' \chi_{KS}(\mathbf{r},\mathbf{r}',\omega) \delta v_{KS}(\mathbf{r}',\omega)$$

$$\delta\rho(\omega) = \chi(\omega) \delta v_{appl}(\omega) \quad \text{notation!}$$

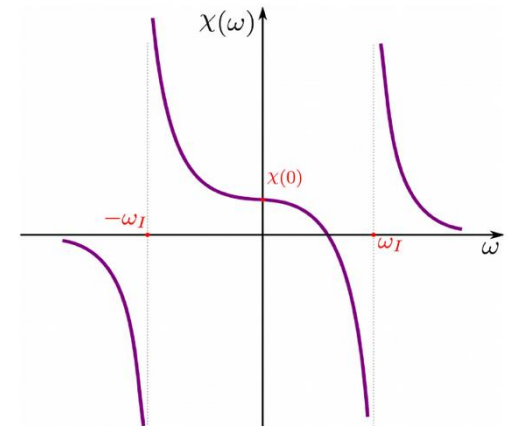
$$\delta v_{KS}(\omega) = \chi_{KS}^{-1}(\omega) \delta\rho(\omega)$$

Where

$$\chi_{KS}(\mathbf{r},\mathbf{r}',\omega) = \sum_{i,a} \frac{\phi_a^*(\mathbf{r})\phi_i(\mathbf{r})\phi_i^*(\mathbf{r}')\phi_a(\mathbf{r}')}{\omega - (\varepsilon_i - \varepsilon_a) + i0^+} - \sum_{i,a} \frac{\phi_i^*(\mathbf{r})\phi_a(\mathbf{r})\phi_a^*(\mathbf{r}')\phi_i(\mathbf{r}')}{\omega + (\varepsilon_i - \varepsilon_a) + i0^+}$$

Note, the poles of χ_{KS} are not the same as those of χ .

$$\omega_{ai} = E_i - E_a \rightarrow \chi_{KS}(\omega) = \infty \rightarrow \delta v_{KS}(\omega_{ai}) = 0$$



Back to de Casida's formulation

$$\delta v_{KS}(\omega) = \chi_{KS}^{-1}(\omega) \delta \rho(\omega)$$

$$\delta v_{KS}(\omega) - \delta v_H(\omega) - \delta v_{XC}(\omega) = \delta v_{appl}(\omega)$$

$$\chi_{KS}^{-1}(\omega) \delta \rho(\omega) - \delta v_H(\omega) - \delta v_{XC}(\omega) = \delta v_{appl}(\omega)$$

$$\chi_{KS}^{-1}(\omega) \delta \rho(\omega) + \frac{-\delta v_H(\omega) - \delta v_{XC}(\omega)}{\delta \rho(\omega)} \delta \rho(\omega) = \delta v_{appl}(\omega)$$

$$\chi_{KS}^{-1}(\omega) \delta \rho(\omega) - \frac{\delta v_H(\omega)}{\delta \rho(\omega)} \delta \rho(\omega) - \frac{\delta v_{XC}(\omega)}{\delta \rho(\omega)} \delta \rho(\omega) = \delta v_{appl}(\omega)$$

$$\chi_{KS}^{-1}(\omega) \delta \rho(\omega) - \frac{1}{|\mathbf{r} - \mathbf{r}'|} \delta \rho(\omega) - f_{XC} \delta \rho(\omega) = \delta v_{appl}(\omega)$$

$$\left[\chi_{KS}^{-1}(\omega) - \frac{1}{|\mathbf{r} - \mathbf{r}'|} - f_{XC} \right] \delta \rho(\omega) = \delta v_{appl}(\omega)$$

Back to de Casida's formulation

$$\left[\chi_{KS}^{-1}(\omega) - \frac{1}{|\mathbf{r} - \mathbf{r}'|} - f_{XC} \right] \delta\rho(\omega) = \delta v_{appl}(\omega)$$

$$\omega_I \rightarrow \chi(\omega_I) = \infty \rightarrow \delta v_{appl} = 0$$

$$\left[\chi_{KS}^{-1}(\omega_I) - \frac{1}{|\mathbf{r} - \mathbf{r}'|} - f_{XC} \right] \delta\rho(\omega) = 0$$

Therefore, at the true excitation energies the operator on the left vanishes

$$\left[\chi_{KS}^{-1}(\omega_I) - \frac{1}{|\mathbf{r} - \mathbf{r}'|} - f_{XC} \right] = 0$$

$$\left[\chi_{KS}(\omega_I) * \chi_{KS}^{-1}(\omega_I) - \chi_{KS}(\omega_I) * \frac{1}{|\mathbf{r} - \mathbf{r}'|} - \chi_{KS}(\omega_I) * f_{xc} \right] = 0$$

$$\left[\hat{1} - \chi_{KS}(\omega_I) * f_{Hxc} \right] = 0$$

root-finding problem



$$\chi_{KS}(\omega) * f_{Hxc} * \zeta(\omega) = \lambda(\omega) \zeta(\omega)$$

$$\text{with } \lambda(\omega_I) = 1$$

Casida's equations

$$\chi_{KS}(\omega) * f_{Hxc} * \zeta(\omega) = \lambda(\omega) \zeta(\omega)$$

with $\lambda(\omega_I) = 1$

By expanding χ_{KS} in its spectral Lehmann representation, Casida managed to cast this cumbersome nonlinear root-finding problem into a single, massive **linear matrix eigenvalue problem** where ω_I is pulled out of the operator and sits cleanly on the right side of the equation as the eigenvalue itself.

$$\begin{pmatrix} \mathbf{A}(\omega) & \mathbf{B}(\omega) \\ \mathbf{B}^*(\omega) & \mathbf{A}^*(\omega) \end{pmatrix} \begin{pmatrix} X(\omega) \\ Y(\omega) \end{pmatrix} = \Omega \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} X(\omega) \\ Y(\omega) \end{pmatrix}$$

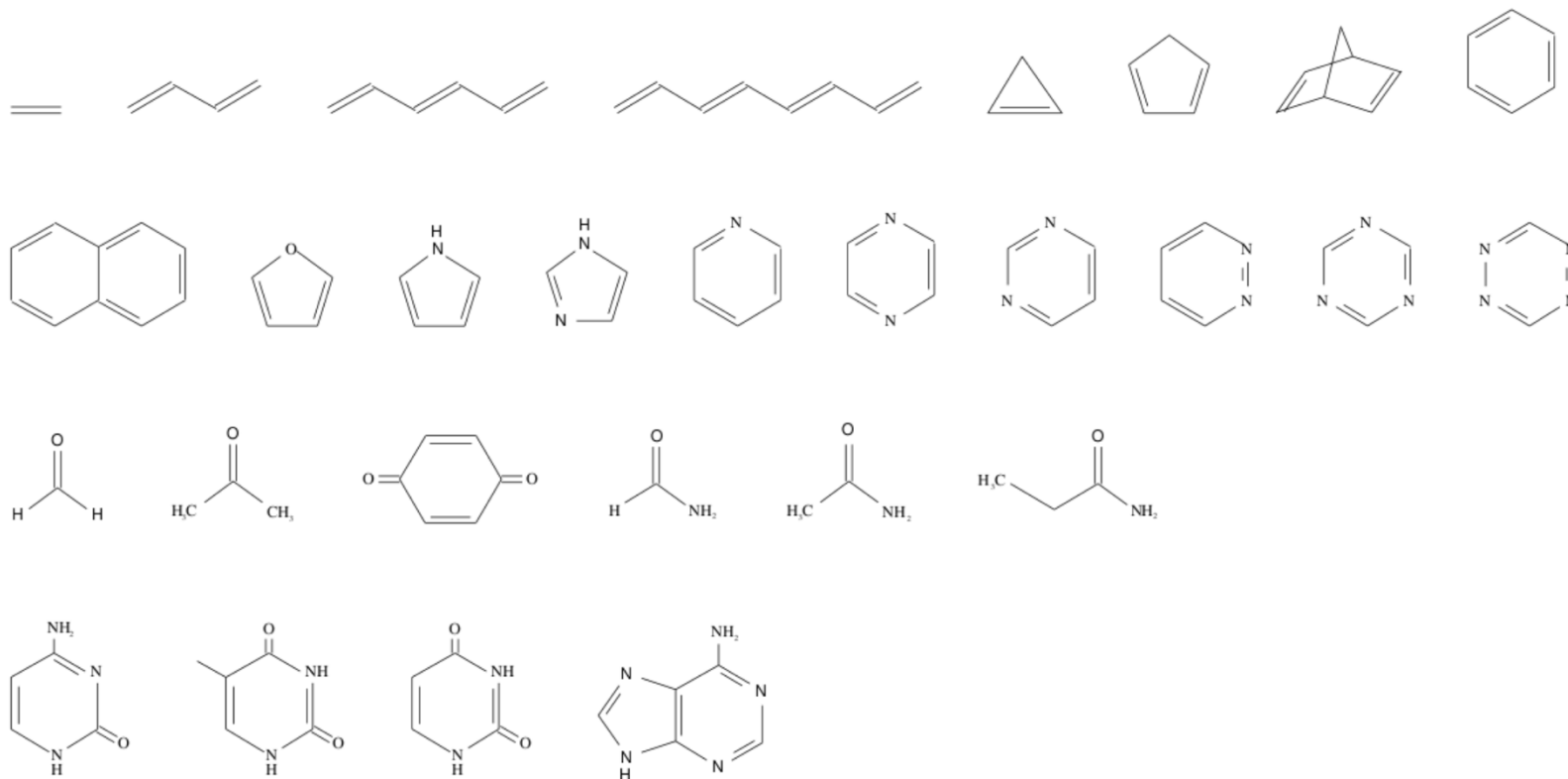
$$(A(\omega))_{ai,bj} = (\varepsilon_a - \varepsilon_i) + \langle ab | \hat{f}_{Hxc} | ij \rangle$$

$$(B(\omega))_{ai,bj} = \langle ii | \hat{f}_{Hxc} | ab \rangle$$

$$a, b = \text{occupied} \quad i, j = \text{unoccupied}$$

Excitation energies are eigenvalues Ω

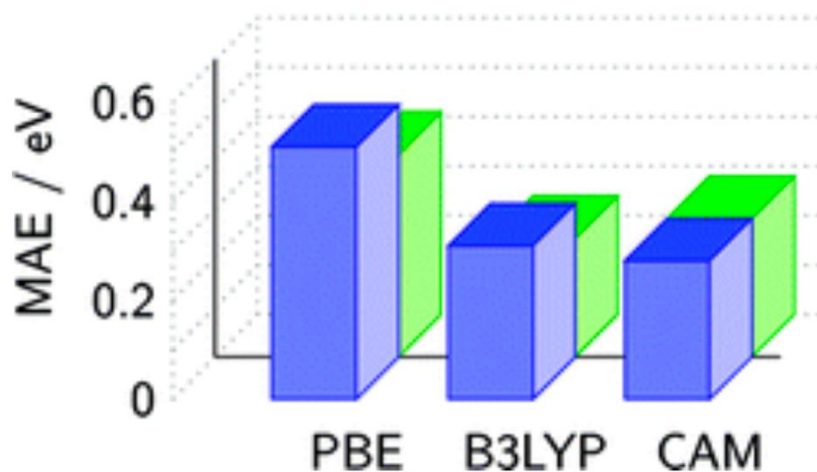
Cual es el mejor funcional xc?



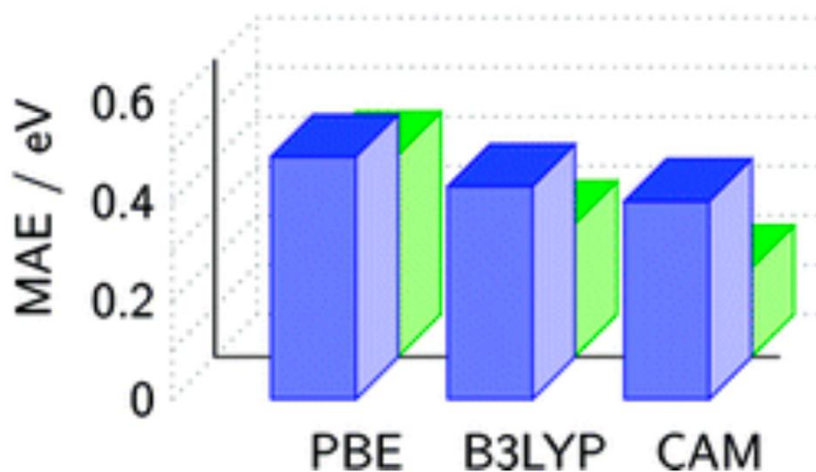
M. Schreiber, M.R. Silva-Junior, S.P.A. Sauer, W. Thiel, Benchmarks for electronically excited states: CASPT2, CC2, CCSD, and CC3, J. Chem. Phys. 128 (2008) 134110

Cual es el mejor funcional xc?

57 Singuletes



63 Tripletos



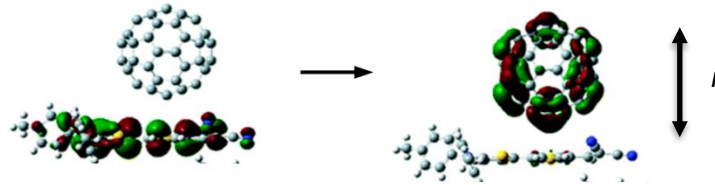
TDDFT completo



Aproximacion Tamm-Dancoff

M. J. G. Peach, M. J. Williamson and D. J. Tozer, Influence of Triplet Instabilities in TDDFT, J. Chem. Theory Comput., 2011, 7, 3578–3585

Excitaciones de transferencia de carga



$$\Delta E_{CT} = EA^A - IP^D - \frac{1}{r}$$

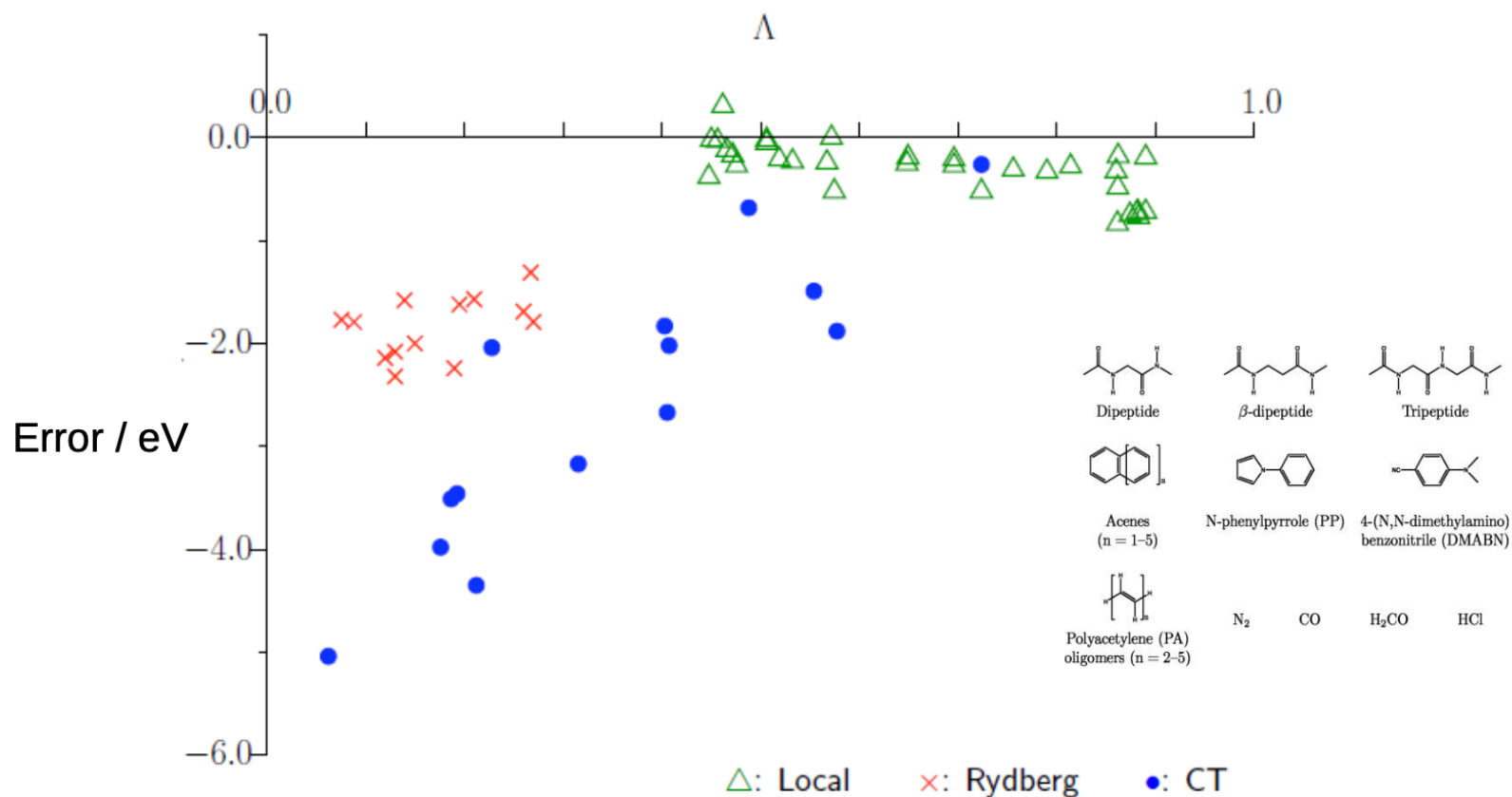
- **Aproximaciones (semi)Locales (ALDA/GGAs):** Estos funcionales asumen que los electrones solo reaccionan a su entorno inmediato: $f_{\chi C}(\mathbf{r}, \mathbf{r}') \propto \delta(\mathbf{r} - \mathbf{r}')$, lo que significa que el kernel se reduce a **cero** si los del donante y el aceptor no se solapan espacialmente.
- **Falta de comportamiento asintótico (1/r):** Al no haber solapamiento entre los orbitales del donante y el aceptor, los elementos de matriz de $f_{\chi C}$ desaparecen. El sistema se comporta como si el electrón transferido no experimentara ninguna fuerza de atracción hacia el hueco que dejó atrás.

$$A_{AD,AD} = (\varepsilon_D - \varepsilon_A) + \langle DD | \hat{f}_{Hxc} | AA \rangle \approx 0$$

$$\Delta E_{CT} \approx \varepsilon_D - \varepsilon_A$$

Excitaciones de transferencia de carga

PBE



Peach, Benfield, Helgaker, Tozer, J. Chem. Phys. 128 044118 (2008)

Hartree-Fock

Energy of a Slater Determinant

$$-\sum_{i=1}^N \frac{1}{2} \nabla_i^2 - \sum_{i=1}^N \sum_{\alpha=1}^M \frac{Z_{\alpha}}{|\mathbf{R}_{\alpha} - \mathbf{r}_i|} + = \sum_{i=1}^N \hat{h}(i) + \sum_i^N \sum_{j>i}^N \frac{1}{r_{ij}}$$

$$\hat{h}(i) = -\frac{1}{2} \nabla_i^2 - \sum_{\alpha=1}^M \frac{Z_{\alpha}}{|\mathbf{R}_{\alpha} - \mathbf{r}_i|} \quad \text{monoelectronic operator}$$

$$\frac{1}{r_{ij}} = \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} \quad \text{bielectronic operator}$$

$$E = \sum_{i=1}^N \langle \phi_i | \hat{h} | \phi_i \rangle + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \langle \phi_i \phi_j | \phi_i \phi_j \rangle - \langle \phi_i \phi_j | \phi_j \phi_i \rangle$$

$$E = \sum_{i=1}^N \langle i | \hat{h} | i \rangle + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \langle ij | ij \rangle - \langle ij | ji \rangle$$

Energy of a Slater Determinant

$$E = \sum_{i=1}^N \langle \phi_i | \hat{h} | \phi_i \rangle + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \langle \phi_i \phi_j | \phi_i \phi_j \rangle - \langle \phi_i \phi_j | \phi_j \phi_i \rangle$$

$$E = \sum_{i=1}^N \langle i | \hat{h} | i \rangle + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \langle ij | ij \rangle - \langle ij | ji \rangle$$

$$\langle i | \hat{h} | i \rangle = \int \phi_i^*(\mathbf{r}_1) \hat{h}(i) \phi_i(\mathbf{r}_1) d\mathbf{r}_1 = h_i$$

$$\langle ij | ij \rangle = \int \phi_i^*(\mathbf{r}_1) \phi_j^*(\mathbf{r}_2) \frac{1}{r_{12}} \phi_i(\mathbf{r}_1) \phi_j(\mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2 \int \left(\langle \sigma_i | \sigma_i \rangle \langle \sigma_j | \sigma_j \rangle \right) = J_{ij}$$

$$\langle ij | ji \rangle = \int \phi_i^*(\mathbf{r}_1) \phi_j^*(\mathbf{r}_2) \frac{1}{r_{12}} \phi_j(\mathbf{r}_1) \phi_i(\mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2 \int \left(\langle \sigma_i | \sigma_j \rangle \langle \sigma_j | \sigma_i \rangle \right) = K_{ij}$$

Meaning of the integrals

$$E = \sum_{i=1}^N \langle i | \hat{h} | i \rangle + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \langle ij | ij \rangle - \langle ij | \textcolor{red}{ji} \rangle$$

$\langle i | \hat{h} | i \rangle = \int \phi_i^*(\mathbf{r}_1) \hat{h}(i) \phi_i(\mathbf{r}_1) d\mathbf{r}_1$ an electron moving in the field of the nuclei

$$J_{ij} = \int \phi_i^*(\mathbf{r}_1) \phi_j^*(\mathbf{r}_2) \hat{h}(i) \phi_i(\mathbf{r}_1) \phi_j(\mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2 \times \underbrace{\left(\langle \sigma_i | \sigma_i \rangle \langle \sigma_j | \sigma_j \rangle \right)}_1$$

$$J_{ij} = \int \left| \phi_i(\mathbf{r}_1) \right|^2 \frac{1}{r_{12}} \left| \phi_j(\mathbf{r}_2) \right|^2 d\mathbf{r}_1 d\mathbf{r}_2 = \int \frac{\rho_i(\mathbf{r}_1) \rho_j(\mathbf{r}_2)}{r_{12}} d\mathbf{r}_1 d\mathbf{r}_2$$

Meaning of the integrals

$$E = \sum_{i=1}^N \langle i | \hat{h} | i \rangle + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \langle ij | ij \rangle - \langle ij | ji \rangle$$

$$\langle ij | ji \rangle = K_{ij} = \int \phi_i^*(\mathbf{r}_1) \phi_j^*(\mathbf{r}_2) \frac{1}{r_{12}} \phi_j(\mathbf{r}_1) \phi_i(\mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2 \times \left(\underbrace{\langle \sigma_i | \sigma_j \rangle}_{\delta_{ij}} \underbrace{\langle \sigma_j | \sigma_j \rangle}_{\delta_{ij}} \right)$$

$$K_{ij} = \begin{cases} \int \phi_i^*(\mathbf{r}_1) \phi_j^*(\mathbf{r}_2) \frac{1}{r_{12}} \phi_j(\mathbf{r}_1) \phi_i(\mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2 & \text{if electrons have the same spin} \\ 0 & \text{if electrons have the different spin} \end{cases}$$

$$K_{ij} < 0$$

Hartree-Fock Equations

$$\delta L = \delta \left[E - \sum_{i,j} \lambda_{ij} (\langle \phi_i | \phi \rangle - \delta_{ij}) \right] = 0$$

$$\delta \left[\sum_{i=1}^N \langle \phi_i | \hat{h} | \phi_i \rangle + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \langle \phi_i \phi_j | \phi_i \phi_j \rangle - \langle \phi_i \phi_j | \phi_j \phi_i \rangle - \sum_{i,j} \lambda_{ij} (\langle \phi_i | \phi \rangle - \delta_{ij}) \right] = 0$$

$$\sum_{i=1}^N \langle \delta \phi_i | \hat{h} | \phi_i \rangle + \langle \phi_i | \hat{h} | \delta \phi_i \rangle + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \langle \delta \phi_i \phi_j | \phi_i \phi_j \rangle + \langle \phi_i \delta \phi_j | \phi_i \phi_j \rangle + \dots = 0$$

$$\hat{f}_i \phi_i(\mathbf{r}) = \varepsilon_i \phi_i(\mathbf{r}) \quad i = 1, 2, \dots, N$$

Hartree-Fock Equations

$$\hat{f}_i \phi_i(\mathbf{r}) = \varepsilon_i \phi_i(\mathbf{r}) \quad i = 1, 2, \dots, N$$

$$\hat{f}_i = \hat{f}(\mathbf{x}_i) = \hat{h}(i) + \sum_{j=1}^N (\hat{J}_j - \hat{K}_j)$$

$$\hat{J}_j = \left\langle \phi_j \left| \frac{1}{r_{12}} \right| \phi_j \right\rangle, \quad \hat{K}_j = f[\phi_i, \phi_j]$$

This is a very complicated integro-differential equation!



Clemens C. J. Roothan

Roothan's equations: SCF

$$\phi_i(\mathbf{r}) = \sum_{\mu=1}^K C_{\mu i} \chi_{\mu}(\mathbf{r})$$

$$\sum_{\nu} F_{\mu\nu} C_{\nu i} = \varepsilon_i \sum_{\nu} S_{\mu\nu} C_{\nu i}$$

$$\mathbf{FC} = \mathbf{SC}\varepsilon$$

$$F_{\mu\nu} = \int \chi_{\mu}^*(\mathbf{x}_1) \hat{f}(\mathbf{x}_1) \chi_{\nu}(\mathbf{x}_1) d\mathbf{x}_1 \quad \text{Fock matrix}$$

$$S_{\mu\nu} = \int \chi_{\mu}^*(\mathbf{x}_1) \chi_{\nu}(\mathbf{x}_1) d\mathbf{x}_1 \quad \text{overlap matrix}$$



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Roothan's equations: SCF

$$\mathbf{FC} = \mathbf{SC}\epsilon$$

$$F_{\mu\nu} = \int \chi_{\mu}^*(\mathbf{x}_1) \hat{f}(\mathbf{x}_1) \chi_{\nu}(\mathbf{x}_1) d\mathbf{x}_1 \quad \text{Fock matrix}$$

$$S_{\mu\nu} = \int \chi_{\mu}^*(\mathbf{x}_1) \chi_{\nu}(\mathbf{x}_1) d\mathbf{x}_1 \quad \text{overlap matrix}$$

1. Specify molecule, basis functions, and electronic state of interest (i.e., singlet, triplet, etc)
2. Form overlap matrix S
3. Guess initial MO coefficients C
4. Form Fock matrix F
5. Solve $\mathbf{FC} = \mathbf{SC}\epsilon$
6. Use new MO coefficients C to build new Fock matrix F
7. Go to step 5; repeat until C no longer changes from one iteration to the next

$$\textit{Scaling } O(N^4)$$